



SECP2753-01 DATA MINING

PROJECT 1

LECTURER : DR ROLIANA BINTI IBRAHIM

NO	NAME	MATRIC NO
1.	PRAVINRAJ A/L SIVABATHI	A23CS0171
2.	DHESHIEGHAN A/L SARAVANA MOORTHY	A23CS0072
3	MIKHAIL BIN YASSIN	A21EC0053

TABLE OF CONTENT

No.	Section Title	Page
1.0	Introduction	3
	1.1 Overview of the dataset	3-4
	1.2 Objective of the analysis	5
	1.3 Business insights to be explored	5
	1.4 Classification models selected for the analysis	6-7
2.0	Preprocessing Process	8
	2.1 Attribute Filtering	8
	2.2 Data Categorization	8-9
	2.3 Data transformation	9
	2.4 Data Export	10
	2.5 Sampling strategy for model training and testing	10
3.0	Classification Models	11
	3.1 Lifestyle Drivers of Obesity	11
	3.1.1 K-Nearest Neighbours (K-NN)	11-12
	3.1.2 Naïve Bayes	12-13
	3.1.3 Decision Tree	13-14
	3.2 At-Risk Demographics	15
	3.2.1 K-Nearest Neighbours (K-NN)	15-16
	3.2.2 Naïve Bayes	17
	3.2.3 Decision Tree	18
4.0	Model Evaluation	19-20
5.0	Conclusion	21
6.0	References	22

1.0 IntroductionFor this assignment, we were tasked with collecting and analyzing a dataset to uncover potential business or healthcare-related insights. We selected an obesity dataset, which captures key health indicators and demographic information of individuals. This dataset is valuable for understanding the relationship between variables such as age, body measurements, eating habits, and physical activity levels with the likelihood of developing obesity, represented by the "Obesity" classification label. Through this analysis, our goal is to extract meaningful patterns using classification models, thereby supporting preventive healthcare decision-making and promoting awareness through predictive analytics.

1.1 Overview of the dataset

The dataset primarily focuses on identifying patterns that influence the likelihood of obesity based on various medical and demographic features. It was sourced from Kaggle, a popular data science platform that hosts numerous open-source datasets. Originally published as part of a cardiovascular health research initiative, this dataset is commonly used for obesity prediction and classification tasks. The dataset contains 17 attributes and 2,111 records, each representing different health-related characteristics of individuals. It is clean and well-structured, with no missing or duplicate values detected. The data includes both numerical and categorical variables, making it suitable for a variety of machine learning and classification algorithms. Below is the list of attributes included in this dataset:

No	Attribute	Description
1	Gender	Gender of the patients
2	Age	Age of the patients
3	Height	In meters
4	Weight	In kgs
5	Family_history	Has a family member suffered or suffers from being overweight?
6	FAVC	Do you eat high caloric food frequently?
7	FCVC	Do you usually eat vegetables in your meals?
8	NCP	How many main meals do you have daily?
9	CAEC	Do you eat any food between meals?
10	SMOKE	Do you smoke?
11	CH2O	How much water do you drink daily?
12	SCC	Do you monitor the calories you eat daily?
13	FAF	How often do you have physical activity?
14	TUE	How much time do you use technological devices such as cell phones, video games, television, computers and others?
15	CALC	How often do you drink alcohol?
16	MTRANS	Which transportation do you usually use?
17	Obesity_level	Obesity level

Table 1.1.1 Data Set List

1.2 Objectives of the Analysis

The primary objective of this analysis is to apply classification models to predict and understand obesity levels among individuals based on their lifestyle choices and demographic characteristics. Specifically, the analysis aims to:

- **Classify individuals by obesity levels**

Predict whether an individual falls into categories such as Normal Weight, Overweight, or Obese using features like eating habits, physical activity, water intake, and frequency of fast food consumption.

- **Assess the influence of demographic factors**

Examine how attributes such as age, gender, and mode of transportation affect obesity risk. These findings can help design more effective and targeted health interventions.

To achieve these objectives, the analysis will apply and compare multiple classification algorithms, including K-Nearest Neighbours (KNN), Naïve Bayes, and Decision Tree models. The goal is to uncover patterns that support preventive healthcare strategies and promote public health awareness.

1.3 Business Insights to Be Explored

Understanding the factors behind obesity can help stakeholders make data-driven decisions in public health, wellness, and policy-making. This analysis will focus on extracting the following actionable insights:

- **Lifestyle Drivers of Obesity**

Identify which lifestyle behaviors such as meal frequency, caloric food consumption, physical activity, and hydration are most strongly associated with obesity.

- **At-Risk Demographics**

Determine which groups (e.g., by gender, age range, or mode of transportation) are more prone to higher obesity levels, enabling more personalized and effective awareness campaigns.

These insights will be especially useful for health organizations, wellness platforms, and policymakers aiming to reduce obesity rates through informed, targeted, and educational interventions.

1.4 Classification models selected for the analysis

To perform the classification tasks outlined above, the following three machine learning models were selected due to their simplicity, interpretability, and effectiveness in handling categorical and numerical data. These models are commonly used for predictive tasks and provide a solid foundation for understanding relationships in healthcare datasets

1.4.1 K-Nearest Neighbours (K-NN)

K-NN is a non-parametric algorithm that classifies data points based on the class of their nearest neighbours in the feature space. It does not make any assumptions about the underlying data distribution and works well with both numerical and categorical data. Its simplicity and intuitive nature make it a good baseline model to start with.

- Advantages: Easy to implement, highly interpretable, and does not require an explicit training phase. It adapts well to datasets with clear boundaries between classes.
- Limitations: Sensitive to irrelevant features and noisy data. The model's performance heavily depends on the choice of distance metric and the value of k (number of neighbours). It may also become computationally expensive with large datasets.

1.4.2 Naïve Bayes

Naïve Bayes is a probabilistic classifier based on Bayes' theorem, with a strong assumption of independence among features. Despite this simplification, it often performs surprisingly well, especially in high-dimensional classification problems and when the feature relationships are not too complex.

- Advantages: Very fast to train and predict, works well with categorical input features, and is effective even with relatively small amounts of training data. It also handles multi-class classification naturally.
- Limitations: Assumes independence among features, which may not always reflect real-world conditions. Performance can degrade if strong correlations exist between variables.

1.4.3 Decision Tree

A decision tree builds a model in the form of a tree-like structure, using branches to represent decision rules and leaf nodes to represent outcomes. It is highly interpretable and easy to visualize, making it ideal for understanding how individual variables affect the prediction.

- Advantages: Easy to understand and explain; capable of handling both numerical and categorical features. It captures non-linear relationships and interactions between variables effectively, often without needing extensive data preprocessing.
- Limitations: Prone to overfitting, especially when the tree is too deep or unpruned. It may also be sensitive to small changes in the data, which can result in different tree structures.

These models will be trained and evaluated across multiple classification targets to determine which performs best for different types of medical prediction tasks. Their performance will be compared using evaluation metrics such as accuracy, precision, recall, and F1-score, which will be discussed in Section 4.

2.0 Data Preparation

```
# Load the dataset
df = pd.read_csv('obesity.csv', delimiter=';')
print("Dataset loaded successfully.")
print("Original shape:", df.shape)

# Clean column names
df.columns = df.columns.str.strip().str.replace(' = ', '_').str.replace(' ', '_')
print("\nCleaned column names:", df.columns.tolist())
```

Figure 2.0.1 Data preparation

All data preprocessing steps for this analysis were conducted using Python, without relying on external tools such as RapidMiner. The dataset was loaded using Pandas, and the initial preparation involved cleaning the column names by removing unnecessary spaces and special characters to ensure consistency and ease of use in subsequent steps. This preprocessing also included identifying and consolidating the one-hot encoded obesity class columns into a single nominal target column. The dataset was confirmed to be clean and well-structured, with no missing or duplicate values, allowing us to proceed directly to transformation and modeling tasks.

2.1 Attribute Filtering

```
# Identify one-hot encoded target columns (Obesity Levels)
obesity_cols = [col for col in df.columns if col.startswith('Obesity_')]
print(f"\nFound {len(obesity_cols)} one-hot encoded obesity columns.")
```

Figure 2.1.1 Attribute filtering

The first step in the preparation process involved reviewing the dataset for any irrelevant or redundant attributes. Unlike some datasets that include ID columns or unrelated metadata, this dataset contained only meaningful attributes directly related to obesity prediction. Therefore, no columns were removed. Additionally, the dataset included several one-hot encoded columns representing obesity levels. These were programmatically identified and consolidated into a single nominal target column for simplified analysis. A check for missing or duplicate values confirmed that the dataset was clean, allowing us to proceed directly to transformation without the need for imputation or data cleansing.

2.2 Data Categorization

```
# Combine one-hot encoded columns into a single nominal column
df['ObesityLevel_nominal'] = df[obesity_cols].idxmax(axis=1)
df['ObesityLevel_nominal'] = df['ObesityLevel_nominal'].str.replace('Obesity_', '')
print("\nCreated a single target column 'ObesityLevel_nominal':")
print(df['ObesityLevel_nominal'].head())
```

Figure 2.2.1 Data Categorization

In this analysis, data categorization was applied to the obesity class labels. The dataset originally represented the target variable using multiple one-hot encoded columns corresponding to each obesity level. To simplify the structure and improve interpretability, these columns were consolidated into a single categorical column named `obesityLevel_nominal`. This transformation provided a unified view of each individual's obesity category, making the data easier to manage for visualization, analysis, and model training.

2.3 Data Transformation

```
# Label encode the nominal obesity target column
label_encoder = LabelEncoder()
df['ObesityLevel_encoded'] = label_encoder.fit_transform(df['ObesityLevel_nominal'])

print("Transformed the single target column into numerical labels:")
print(df[['ObesityLevel_nominal', 'ObesityLevel_encoded']].head())

print("\nMapping of encoded values to original labels:")
for i, class_name in enumerate(label_encoder.classes_):
    print(f"{i}: {class_name}")

# Define features and target
X = df.drop(columns=obesity_cols + ['ObesityLevel_nominal', 'ObesityLevel_encoded'])
y = df['ObesityLevel_encoded']
```

Figure 2.3.1 Data Transformation

After categorizing the obesity class labels into a single nominal column, the next step involved transforming this categorical data into a numerical format suitable for machine learning algorithms. Most algorithms require numerical inputs, and thus the `ObesityLevel_nominal` column was converted into integer labels using label encoding. This transformation was performed using Scikit-learn's `LabelEncoder`, which assigned a unique integer to each obesity category. The resulting column, `ObesityLevel_encoded`, served as the final target variable (`y`) for classification modeling. This encoding preserved the categorical nature of the labels while making them compatible with the training and evaluation process.

2.4 Data Export

```
# Prepare final dataset and export to CSV
df_final_for_export = X.copy()
df_final_for_export['ObesityLevel'] = y
df_final_for_export.to_csv('obesity_cleaned_and_restructured.csv', index=False)
print("Cleaned data has been exported to 'obesity_cleaned_and_restructured.csv'")
```

Figure 2.4.1 Data Export

Once all transformations were completed, the final version of the dataset consisting of the selected feature attributes and the encoded target variable was saved into a new CSV file using Python's `to_csv()` function. The file, named `obesity_cleaned_and_restructured.csv`, contained all relevant data in a clean and structured format. This exported dataset served as the primary input for training and testing the classification models in the subsequent stages of analysis. Exporting the data also ensured that the preprocessing steps could be reused or shared without repeating the entire pipeline.

2.5 Sampling strategy for model training and testing

```
# 70-30 train-test split with stratification
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.30,
    random_state=42,
    stratify=y
)

print(f"Data split into training and testing sets.")
print(f"Features (X) shape: {X.shape}")
print(f"Target (y) shape: {y.shape}")
print("-" * 20)
print(f"Training set shape (X_train): {X_train.shape}")
print(f"Testing set shape (X_test): {X_test.shape}")
```

Figure 2.5.1 Sampling Strategy

To ensure fair and accurate model evaluation, the dataset was divided into training and testing subsets using a 70:30 split. Specifically, 70% of the data was allocated for training, while the remaining 30% was reserved for testing the model's performance on unseen data. This splitting was carried out using Scikit-learn's `train_test_split()` function, with stratification applied based on the target variable to maintain class distribution across both sets. Stratification helps ensure that the proportion of each obesity category remains consistent, improving the reliability of model evaluation. The training data was used to learn the relationships between features and obesity levels, while the test data provided an unbiased measure of generalization performance.

3.0 Classification Models

This section evaluates three machine learning classification models to extract actionable insights related to obesity prediction: K-Nearest Neighbours (KNN), Naïve Bayes, and Decision Tree. These models are trained and tested on the pre-processed dataset using a 70:30 split, as established earlier. Model performance is measured using standard evaluation metrics, including accuracy, precision, recall, and F1-score, to determine how effectively each model predicts obesity categories based on lifestyle and demographic features.

3.1 Lifestyle Drivers of Obesity

Lifestyle habits play a significant role in determining obesity levels. By analyzing variables such as number of meals per day, fast food consumption, water intake, and physical activity, the models help identify which daily behaviors are most strongly associated with higher obesity risk. These insights support the development of data-driven wellness strategies aimed at modifying specific lifestyle factors to prevent or reduce obesity.

3.1.1 K-Nearest Neighbours (K-NN)

```
# Scale Data and Train KNN Model |
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train_scaled, y_train)
y_pred = knn_model.predict(X_test_scaled)
```

Figure 3.1.1.1 Dataset using K-NN

	true Insufficient_Weight	true Normal_Weight	true Overweight_Level_I	true Overweight_Level_II	true Type_I	true Type_II	true Type_III	class precision
pred. Insufficient_Weight	75	21	6	3	1	2	0	69.44%
pred. Normal_Weight	5	36	8	4	3	0	1	63.16%
pred. Overweight_Level_I	2	13	54	9	8	1	0	62.07%
pred. Overweight_Level_II	0	9	11	52	7	1	1	64.20%
pred. Type_I	0	5	7	5	80	2	0	80.81%
pred. Type_II	0	2	1	14	6	83	0	78.30%
pred. Type_III	0	0	0	0	1	0	95	98.96%
class recall	91.46%	41.86%	62.07%	59.77%	75.47%	93.26%	97.94%	

Figure 3.1.1.2 Result of K-NN

The KNN classification model for predicting Obesity Level shows a respectable overall accuracy of 74.92%, correctly classifying about three-quarters of the individuals in the test set. Precision is strongest for the more distinct categories, particularly 'Type_III' with 98.96% and 'Type_I' with 80.81%, meaning predictions for these groups are highly reliable. Precision is weaker for the middle categories like 'Normal_Weight' (63.16%) and 'Overweight_Level_I' (62.07%), indicating a higher rate of misclassification for these groups. In terms of recall, the model excels at identifying true cases of 'Type_III' (97.94%) and 'Type_II' (93.26%). However, its primary weakness lies in identifying individuals of 'Normal_Weight', with a recall of only 41.86%, suggesting that a majority of true 'Normal_Weight' individuals are misclassified as something else. The F1-score would be high for the obesity types but poor for 'Normal_Weight', indicating the model effectively identifies clear cases of obesity but struggles to differentiate between the more nuanced 'Normal' and 'Overweight' classes.

3.1.2 Naïve Bayes

```
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report

# Train Naïve Bayes model
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)

# Predict and evaluate
y_pred = nb_model.predict(X_test)
print("Naïve Bayes Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, target_names=label_encoder.classes_))
```

Figure 3.1.2.1 Dataset using Naive Bayes

Overall Accuracy: 53.94%

	true Insufficient_Weight	true Normal_Weight	true Overweight_Level_I	true Overweight_Level_II	true Type_I	true Type_II	true Type_III	class precision
pred. Insufficient_Weight	78	53	21	16	0	0	0	46.43%
pred. Normal_Weight	0	17	7	1	1	0	0	65.38%
pred. Overweight_Level_I	3	7	13	1	2	0	0	50.00%
pred. Overweight_Level_II	0	3	2	16	3	2	1	59.26%
pred. Type_I	0	3	20	13	37	1	1	49.33%
pred. Type_II	1	3	24	40	62	86	0	39.81%
pred. Type_III	0	0	0	0	1	0	95	98.96%
class recall	95.12%	19.77%	14.94%	18.39%	34.91%	96.63%	97.94%	

Figure 3.1.2.2 Result of Naive Bayes

In the Naïve Bayes classification model, the overall accuracy is 53.94%, indicating the model correctly classifies just over half of the instances. The model's performance is highly polarized. 'Type_III' has the highest precision at 98.96%, but other classes like 'Type_II' suffer from very low precision (39.81%), meaning these predictions are often incorrect. The model's recall tells a similar story of imbalance. It is very effective at identifying true instances of 'Type_III' (97.94%), 'Type_II' (96.63%), and 'Insufficient_Weight' (95.12%). However, it performs extremely poorly on the central weight categories, with recall scores of only 19.77% for 'Normal_Weight' and 14.94% for 'Overweight_Level_I'. This extreme disparity between high recall for some classes and very low recall for others indicates a strong model bias. The F1-score would be very low for the 'Normal' and 'Overweight' classes, suggesting the model has learned to identify the extremes but fails to capture the nuances of the intermediate weight statuses.

3.1.3 Decision Tree

```
# Train Decision Tree model
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)

# Predict and evaluate
y_pred = dt_model.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, target_names=label_encoder.classes_))
```

Figure 3.1.3.1 Dataset using Decision Tree

Overall Accuracy: 92.43%

	true Insufficient_Weight	true Normal_Weight	true Overweight_Level_I	true Overweight_Level_II	true Type_I	true Type_II	true Type_III	class precision
pred. Insufficient_Weight	74	4	0	0	0	0	0	94.87%
pred. Normal_Weight	8	73	5	0	0	0	0	84.88%
pred. Overweight_Level_I	0	8	74	1	0	0	0	89.16%
pred. Overweight_Level_II	0	1	8	80	2	0	0	87.91%
pred. Type_I	0	0	0	5	102	1	0	94.44%
pred. Type_II	0	0	0	1	2	87	1	95.60%
pred. Type_III	0	0	0	0	0	1	96	98.97%
class recall	90.24%	84.88%	85.06%	91.95%	96.23%	97.75%	98.97%	

Figure 3.1.3.2 Result of Decision tree

The Decision Tree model demonstrates exceptional performance, achieving a high overall accuracy of 92.43%. This model is both accurate and well-balanced. Precision is consistently high across all seven categories, ranging from a strong 84.88% for 'Normal_Weight' to an almost perfect 98.97% for 'Type_III'. This means its predictions are highly reliable, regardless of the class. Similarly, recall is excellent across the board, with most classes achieving over 84%. The model correctly identifies 98.97% of 'Type_III' individuals and 97.75% of 'Type_II'. Because both precision and recall are strong and balanced for every category, the corresponding F1-scores would also be very high. This indicates that the Decision Tree model is the most robust and effective classifier for this task, successfully learning the patterns that distinguish between all seven obesity levels without significant bias.

3.2 At-Risk Demographics

This section aims to identify which demographic groups are more susceptible to higher obesity levels based on the analysis. By examining variables such as gender, age range, and mode of transportation, we can uncover patterns that reveal which populations are at greater risk. These insights are crucial for designing personalized health awareness campaigns and targeted intervention strategies, enabling policymakers and health organizations to focus their efforts on the most vulnerable groups. For instance, if a particular age group or transport behavior is linked with higher obesity prevalence, wellness programs can be adapted to address the specific habits and challenges of that group.

3.2.1 K-Nearest Neighbours (K-NN)

```
# Prepare target variable (Age Group)
df_age_task['AgeGroup_nominal'] = df_age_task[['Young', 'Adult', 'Senior']].idxmax(axis=1)
y = LabelEncoder().fit_transform(df_age_task['AgeGroup_nominal'])

# Drop age & obesity columns to avoid bias
X = df_age_task.drop(columns=['Young', 'Adult', 'Senior'] + obesity_cols + ['AgeGroup_nominal'])

# Split and scale data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, stratify=y, random_state=42)
X_train_scaled = StandardScaler().fit_transform(X_train)
X_test_scaled = StandardScaler().fit(X_train).transform(X_test)

# Train and evaluate KNN model
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train_scaled, y_train)
y_pred = knn_model.predict(X_test_scaled)
```

Figure 3.2.1.1 Dataset using K-NN

	true Adult	true Senior	true Young	class precision
pred. Adult	88	3	17	81.48%
pred. Senior	1	1	0	50.00%
pred. Young	23	0	501	95.61%
class recall	78.57%	25.00%	96.72%	

Figure 3.2.1.2 Result of K-NN

In the KNN classification model for Age Group, the overall accuracy is an impressive 93.06%, indicating that the model correctly predicts the age group in the vast majority of cases. The model shows excellent precision for the 'Young' group at 95.61% and strong precision for 'Adult' at 81.48%. This means that when the model identifies an individual as 'Young' or 'Adult', it is very likely to be correct. The precision for the 'Senior' group is 50.00%; however, this is based on only two predictions, making it statistically less significant. In terms of recall, the model excels at finding true 'Young' individuals (96.72%) and performs well for 'Adults' (78.57%). Its primary weakness is the 'Senior' class, where it only identifies 25.00% of the actual seniors, showing it struggles with this underrepresented group. Overall, the KNN model is highly effective for the dominant 'Young' and 'Adult' classes but is unreliable for identifying the 'Senior' demographic due to the class imbalance.

3.2.2 Naïve Bayes

```
# Create target: Age group from one-hot encoded columns
df_age_task['AgeGroup_nominal'] = df_age_task[['Young', 'Adult', 'Senior']].idxmax(axis=1)
y = LabelEncoder().fit_transform(df_age_task['AgeGroup_nominal'])

# Drop irrelevant columns (age and obesity info)
X = df_age_task.drop(columns=['Young', 'Adult', 'Senior'] + obesity_cols + ['AgeGroup_nominal'])

# Train/test split (70:30)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, stratify=y, random_state=42)

# Train and predict using Gaussian Naïve Bayes
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)
y_pred = nb_model.predict(X_test)
```

Figure 3.2.21 Dataset using Naive Bayes

	true Adult	true Senior	true Young	class precision
pred. Adult	90	2	319	21.90%
pred. Senior	16	2	84	1.96%
pred. Young	6	0	115	95.04%
class recall	80.36%	50.00%	22.20%	

Figure 3.2.2.2 Result of Naive Bayes

The Naïve Bayes classification model for Age Group performs poorly, with an overall accuracy of only 32.65%. The model exhibits a paradoxical performance: while it has very high precision for the 'Young' class (95.04%), its recall for this same group is a dismal 22.20%. This suggests that while the model is very confident in the few 'Young' predictions it makes, it incorrectly classifies most young people into other categories. Conversely, the recall for the 'Adult' class is high at 80.36%, but this is achieved through a very low precision of 21.90%, meaning it over-predicts the 'Adult' category and misclassifies a large number of 'Young' individuals. The model completely fails on the 'Senior' class, with a precision of just 1.96%. The F1-score, which balances precision and recall, would be extremely low for all classes except 'Young', highlighting the model's severe bias and ineffectiveness for this classification task.

3.2.3 Decision Tree

```
# Create target from one-hot encoded age columns
df_age_task['AgeGroup_nominal'] = df_age_task[['Young', 'Adult', 'Senior']].idxmax(axis=1)
y = LabelEncoder().fit_transform(df_age_task['AgeGroup_nominal'])

# Drop irrelevant columns (age + obesity info)
X = df_age_task.drop(columns=['Young', 'Adult', 'Senior'] + obesity_cols + ['AgeGroup_nominal'])

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, stratify=y, random_state=42)

# Train and predict using Decision Tree
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
y_pred = dt_model.predict(X_test)
```

Figure 3.2.3.1 Dataset using Decision Tree

Overall Accuracy: 93.53%

	true Adult	true Senior	true Young	class precision
pred. Adult	95	3	18	81.90%
pred. Senior	0	0	2	0.00%
pred. Young	17	1	498	96.51%
class recall	84.82%	0.00%	96.14%	

Figure 3.2.3.2 Result of Decision Tree

The Decision Tree model is the top performer for Age Group classification, achieving an outstanding overall accuracy of 93.53%. Its strength lies in its highly accurate predictions for the majority classes, with a precision of 96.51% for 'Young' and 81.90% for 'Adult'**. This indicates that its classifications for these groups are very trustworthy. The model's recall is also excellent for Young(96.14%) and Adult(84.82%), showing it correctly identifies the vast majority of individuals in these demographics. However, the model completely fails to identify the Senior'` class, recording 0.00% for both precision and recall. The Decision Tree has learned that it can achieve higher overall accuracy by ignoring the tiny 'Senior' minority class and focusing on the dominant groups. While it is the best model for profiling 'Young' and 'Adult' individuals, it is entirely unsuitable for identifying seniors.

4.0 Model Evaluation

In this section, we observe, compare, and evaluate each key business insight: lifestyle drivers of obesity and at-risk demographics. Our analysis is based on the performance metrics (accuracy, precision, and recall) from the three classification models (K-Nearest Neighbour, Naïve Bayes, and Decision Tree). Through this evaluation, we can determine which model best fits each business insight, allowing us to identify key variables to improve public health strategies.

4.1 Lifestyle Drivers of Obesity (Predicting Obesity Level)

This analysis aimed to predict an individual's obesity category based on their lifestyle and physical attributes. The Decision Tree model was the clear winner.

Model	Accuracy	Strengths	Weaknesses
K-Nearest Neighbour	74.92%	Good recall for extreme categories like Type_II (93.26%) and Type_III (97.94%).	Very poor recall for Normal_Weight (41.86%), misclassifying the majority of healthy individuals.
Naïve Bayes	53.94%	Excellent recall for Insufficient_Weight and obesity types II & III.	Extremely biased. Very low recall for Normal_Weight (19.77%) and Overweight categories, making it unreliable for a full population view.
Decision Tree	92.43%	Best Performer. Achieved high, balanced precision and recall across all 7 classes. Excellent at distinguishing between nuanced categories.	Minimal weaknesses; slight confusion between adjacent categories (e.g., Normal_Weight vs. Overweight_Level_I).

Table 4.1.1 Performance metrics of KNN, Naïve Bayes, and Decision Tree models across

The Decision Tree model is unequivocally the best classifier for predicting obesity levels. With an accuracy of 92.43% and consistently high precision and recall (most above 84%), it provides a reliable and balanced view across all seven weight categories. Unlike KNN and Naïve Bayes, which struggled to differentiate the middle classes (Normal_Weight, Overweight), the Decision Tree successfully learned the complex patterns distinguishing each group. This makes it the ideal model for identifying the specific lifestyle factors that drive different obesity outcomes.

4.2 At-Risk Demographics (Predicting Age Group)

This analysis aimed to create demographic profiles by predicting an individual's age group based on their health and lifestyle data. The Decision Tree model again emerged as the top performer, though with a significant caveat.

Model	Accuracy	Strengths	Weaknesses
K-Nearest Neighbour	93.06%	Excellent performance on the two majority classes, Young (96.72% recall) and Adult (78.57% recall).	Poor performance on the minority Senior class (25% recall).
Naïve Bayes	32.65%	Unsuitable for this task.	Severe misclassification issues, with extremely low precision for Senior (1.96%) and low recall for Young (22.20%).
Decision Tree	93.53%	Best Performer. Highest accuracy. Extremely high precision and recall for the Young class.	Completely fails on the Senior class (0% recall). The model overfits the majority classes and has learned to ignore the minority.

Table 4.2.1 Performance metrics of KNN, Naïve Bayes, and Decision Tree models

The Decision Tree model achieved the highest accuracy (93.53%) for profiling age groups. It is exceptionally good at identifying the characteristics of the 'Young' and 'Adult' demographics. However, both the Decision Tree and KNN models highlight a critical limitation: they fail to reliably identify the 'Senior' population. The Decision Tree's 0% recall for seniors means it is unusable for understanding this specific at-risk group. Therefore, while the Decision Tree is the best model for creating profiles of young and adult individuals, a different approach (such as gathering more data on seniors or using advanced sampling techniques) would be needed to profile the senior demographic.

5.0 Conclusion

In this project, we assessed three classification models: K-Nearest Neighbours (KNN), Naïve Bayes, and Decision Tree. The analysis focused on two primary objectives: identifying lifestyle drivers of obesity and profiling at-risk demographic groups by age.

The main conclusions are as follows:

1. Lifestyle Drivers of Obesity Prediction

The Decision Tree model was the superior performer, with an outstanding accuracy of 92.43%. Its balanced precision-recall ratios across all seven obesity levels make it the most reliable tool for understanding which factors (Height, Weight, FAVC, etc.) are the strongest predictors of an individual's weight status. This model can be trusted to inform evidence-based public health interventions.

2. At-Risk Demographics (Age Group) Prediction

The Decision Tree model again achieved the highest accuracy (93.53%), making it the best choice for profiling the dominant 'Young' and 'Adult' demographics. Its high precision and recall for these groups allow for the confident creation of age-specific health profiles. However, all models, including the Decision Tree, completely failed to identify the 'Senior' group, highlighting a significant data imbalance issue that must be addressed in future work.

Based on this evaluation, we suggest a targeted strategy for leveraging these insights:

- The Decision Tree model from the lifestyle analysis (3.1) should be used to identify and rank the most critical risk factors for obesity. Health campaigns should prioritize addressing these specific behaviors (e.g., promoting physical activity, discouraging high-calorie food consumption).
- The Decision Tree model from the demographic analysis (3.2) can be used to create detailed profiles for 'Young' and 'Adult' populations. This allows for the design of age-appropriate health messaging and interventions.
- The failure to model the 'Senior' demographic is an insight in itself. It signals a need for further data collection focused on this group or the application of advanced data science techniques like SMOTE (Synthetic Minority Over-sampling Technique) to create a more balanced dataset for analysis.

In conclusion, by selecting the best-performing model for each specific analytical question, this project provides a clear, data-driven framework. The Decision Tree model offers powerful and reliable insights into both the causes of obesity and the profiles of key demographics, enabling health organizations to create more effective and targeted strategies to promote public well-being.

Dataset Link :

<https://www.kaggle.com/datasets/ruchikakumbhar/obesity-prediction>

6.0 References

1. Ibrahim, R. (2023). Data Preparation using RapidMiner [Lecture notes]. SECPH 2753: Data Mining. Universiti Teknologi Malaysia, Faculty of Computing.
2. **IBM Cloud Education** (2020). *K-Nearest Neighbors (KNN) Algorithm for Machine Learning*. IBM.
<https://www.ibm.com/cloud/learn/k-nearest-neighbors>
3. **Scikit-learn Documentation** (2023). *Naïve Bayes*.
scikit-learn.
https://scikit-learn.org/stable/modules/naive_bayes.html
4. **Google Developers** (2021). *Decision Trees: How to Optimize for Business Decisions*. Machine Learning Crash Course.
<https://developers.google.com/machine-learning/decision-forests/decision-trees>

